

Homework (Carolina Reaper)

- Q1.** What is the greatest common factor of $6x^2$ and $9x^3$?
(A) 3
(B) $3x$
(C) $3x^2$
(D) $6x$
(E) $9x^2$
- Q2.** Simplify the expression $4x^2 + 8x$ by factoring out the greatest common factor.
(A) $4(x + 2)$
(B) $4x(x + 2)$
(C) $2(2x^2 + 4x)$
(D) $4x(x^2 + 2)$
(E) $8(x^2 + x)$
- Q3.** A painter has two canvases with dimensions $12x$ and $15x$. What is the greatest common factor of their areas?
(A) $3x$
(B) $3x^2$
(C) $12x$
(D) $15x$
(E) $60x^2$
- Q4.** A musician plays a rhythm pattern with 12 beats and another with 18 beats. What is the greatest common factor of the number of beats?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12
- Q5.** Simplify the expression $24x^3y + 36x^2y$ by factoring out the greatest common factor.
(A) $12xy(2x^2 + 3x)$
(B) $12x^2y(2x + 3)$
(C) $12x^2y(2x + 3)$
(D) $24x^2y(x + 3)$
(E) $36xy(x^2 + 2)$
- Q6.** What is the greatest common factor of $50x$ and $70y$?
(A) 10
(B) $10x$
(C) $10y$
(D) $10xy$
(E) $10y^2$
- Q7.** Simplify the expression $20x^2y^2 + 30x^2y$ by factoring out the greatest common factor.
(A) $10x^2y(2y + 3)$
(B) $10x^2y(2y + 3)$
(C) $20x^2y(y + 3)$
(D) $20xy^2(2x + 3)$
(E) $30xy(x^2 + 2)$
- Q8.** A sound engineer mixes two audio tracks with frequencies 24 Hz and 30 Hz. What is the greatest common factor of the frequencies?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 24

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $18x^3y + 24x^2y + 30xy$ by factoring out the greatest common factor.
- Q10.** A graphic designer creates two logos with dimensions $15x$ by $20x$ and $18x$ by $25x$. What is the greatest common factor of their areas? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to identify the greatest common factor by listing the factors of each term.
- Tip 2: Remind students to simplify expressions by factoring out the greatest common factor.
- Tip 3: Emphasize the importance of showing work and explaining reasoning for free-response questions.